

Data Mining

Classification: Part 1

Mohammed Brahimi & Sami Belkacem

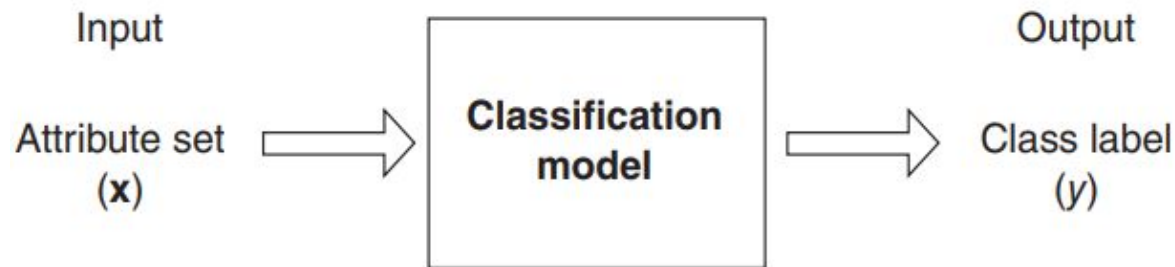
Chapter Overview

- ❑ Introduction to Classification
- ❑ Decision Tree Induction
 - ❑ Introduction to Decision Trees
 - ❑ Attribute Test Conditions
 - ❑ Impurity Measures and Splitting Strategies
 - ❑ Gain Ratio

Chapter Overview

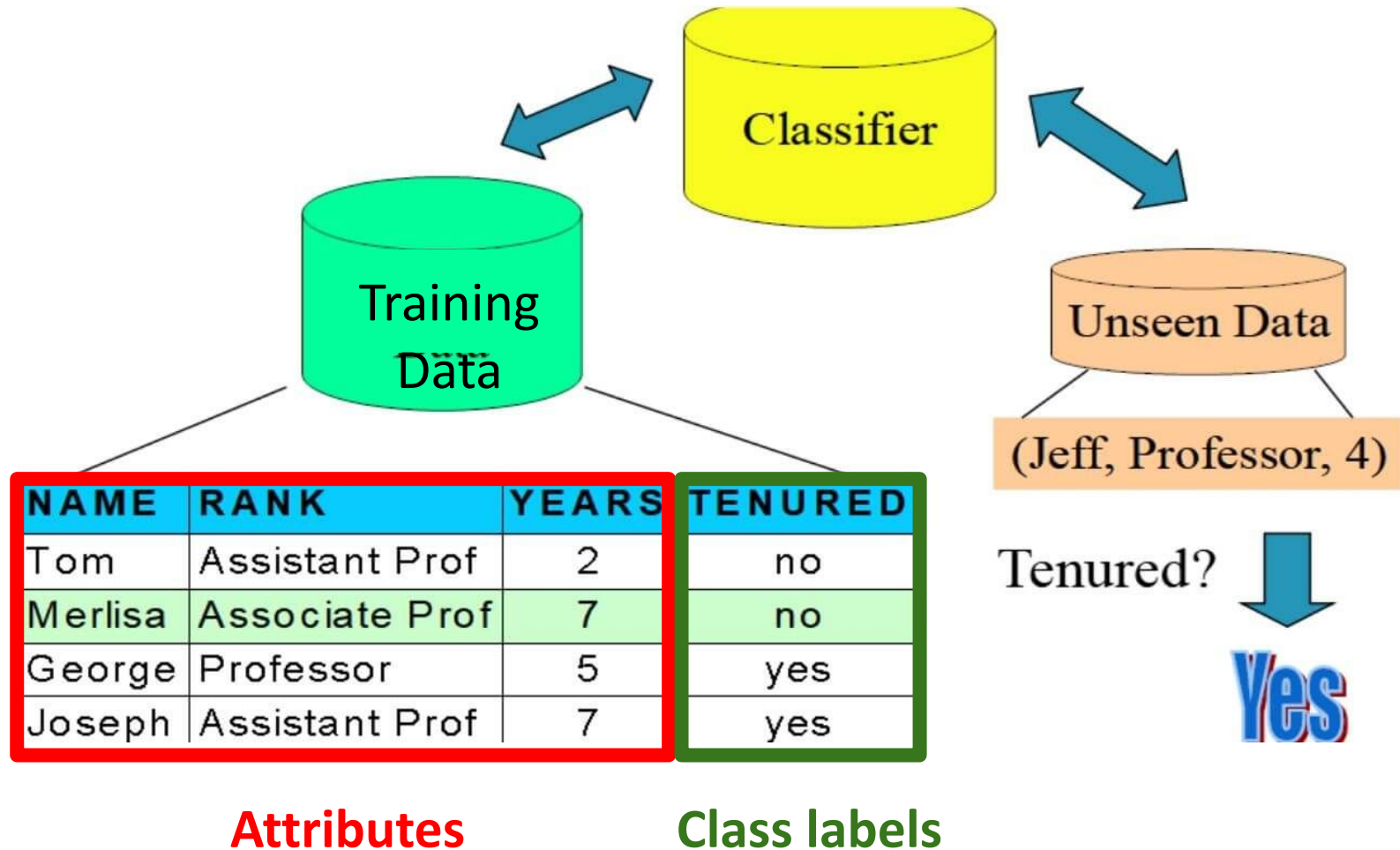
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What is Classification?



- **Dataset**
 - **Attributes:** Descriptive features of instances (x)
 - **Class labels:** Categorical labels representing the class of an instance (Y)
- **Classification**
 - Assigning class labels to instances based on attributes.
- **Classification model**
 - A function $f(x)$ used to classify unseen data x by assigning a class label Y .

Example of Classification



Applications of Classification

- **Healthcare**
 - Medical diagnosis
 - Patient risk categorization
- **Computer Security**
 - Spam filtering
 - Malware classification
- **Banking and Finance**
 - Loan default prediction
 - Credit card fraud detection
- **Retail and E-commerce**
 - Customer purchase pattern classification
 - Sentiment analysis from customer reviews
- **And many more ...**

Role of Classification Models

Predictive Models

- Used to predict class labels for new, unseen data.
- Learn patterns from historical data to make predictions.

Example:

Classify emails as 'urgent' or 'non-urgent' based on the content.



Descriptive Models

- Help understand distinguishing features of different classes.
- Analyzes the data to find common characteristics and patterns.

Example:

Identify factors that differentiate high-risk patients in healthcare.



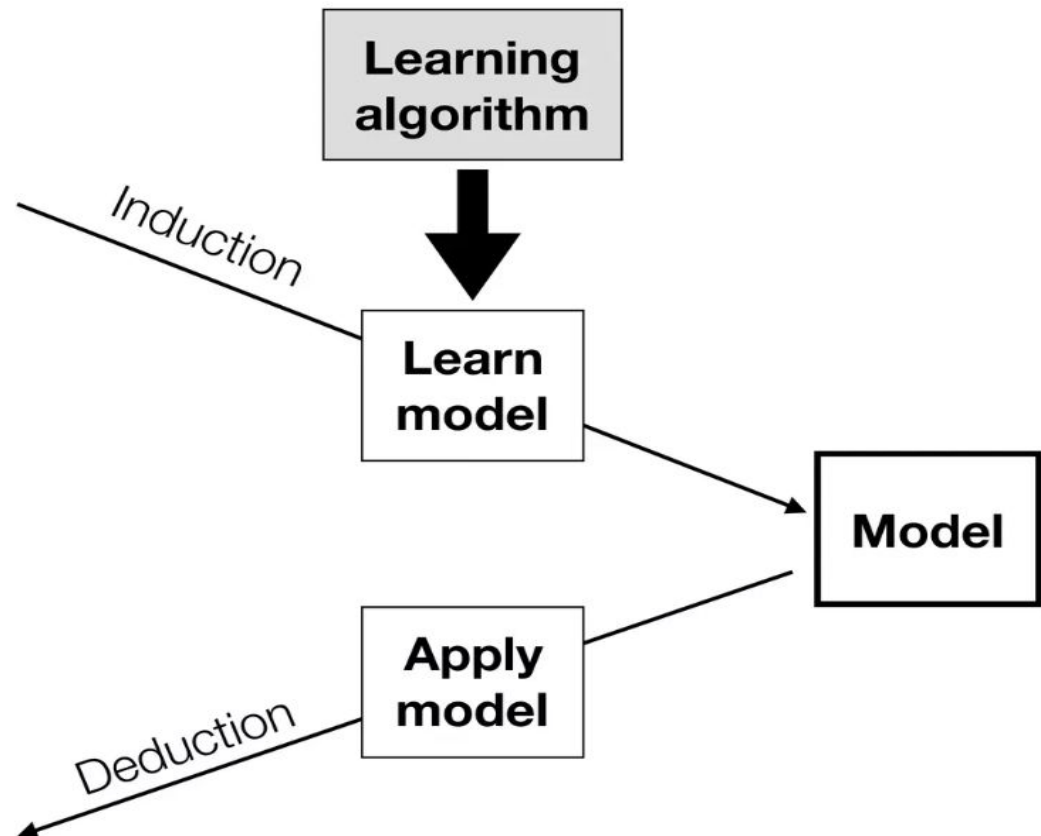
General Framework for Classification

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



General Framework for Classification

- **Induction (Training)**

- Learn a model from a labeled training set.
- Use learning algorithm to build models.

- **Deduction (Testing)**

- Apply the learned model to new unseen instances to predict class labels.
- Assess the model's performance to measure its generalization capability.

The feedback from testing is often used to refine and improve the training process.

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Decision Tree Induction

Definition:

The process of building a decision tree from labeled training data to predict outcomes.

Goal:

Find optimal splits in data to form a tree that separates classes as accurately as possible.

Key steps:

Select attribute: Choose an attribute to split data based on a criterion; e.g. Entropy

Split data: Partition data into subsets based on attribute values.

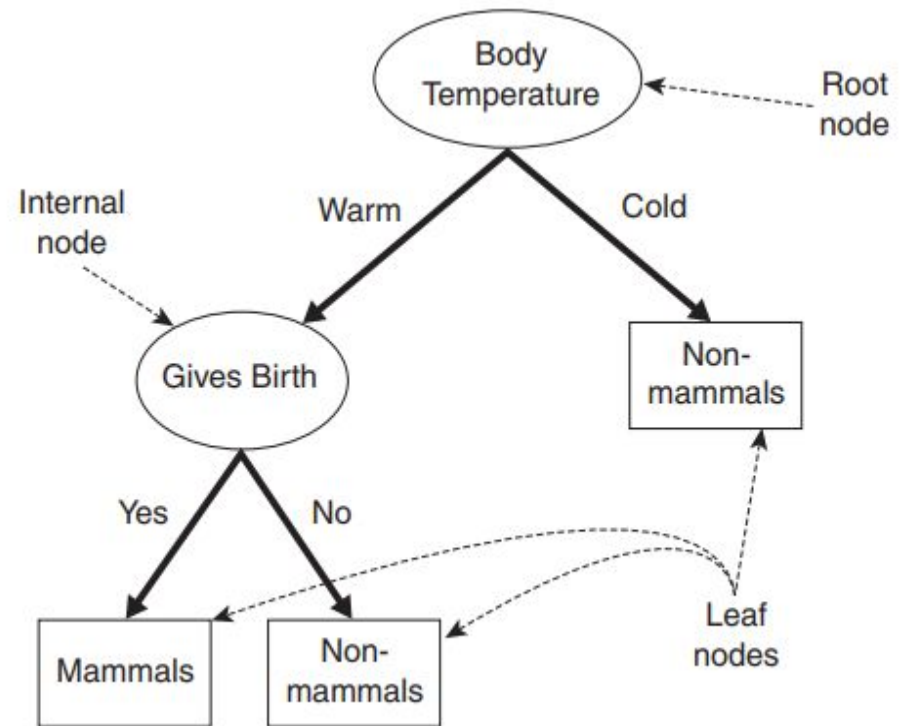
Repeat: Continue splitting until stopping criteria are met

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Decision Tree

- **Root Node**
Initial point of decision-making.
- **Internal Nodes**
Question based on a single attribute.
- **Leaf Nodes**
The final classification outcome.



Mammal classification tree

Deduction in Decision Trees

- **Start at Root Node**

Apply initial attribute test.

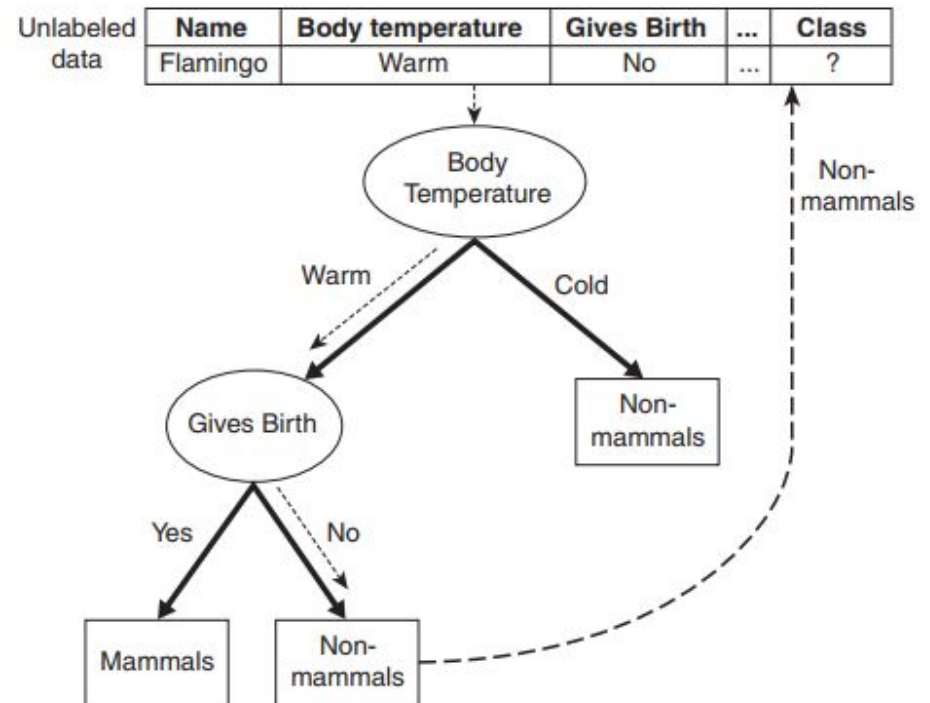
Follow test outcome.

- **Visit next node**

Follow test outcome.

- **Reach Leaf Node**

Determine final classification.



Mammal classification tree

Hunt's Algorithm for Decision Tree building

- **Initial Node**

- Start with a root node containing all instances.

All instances

- **Expansion and Child Nodes Formation**

- Expand nodes with **mixed class instances**.
- Select the attribute test based on a **splitting criterion**.
- Create child nodes for each test outcome
- Distribute instances accordingly.

- **Recursive Process**

- Continue expansion for nodes with mixed class instances.

- **Termination**

- Stop when a node has instances of only one class.

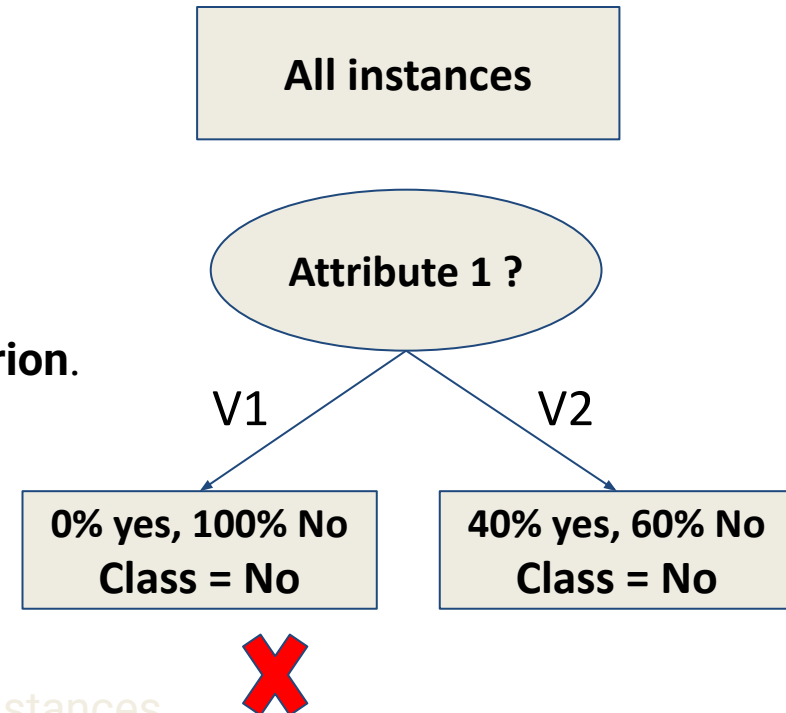
Hunt's Algorithm for Decision Tree building

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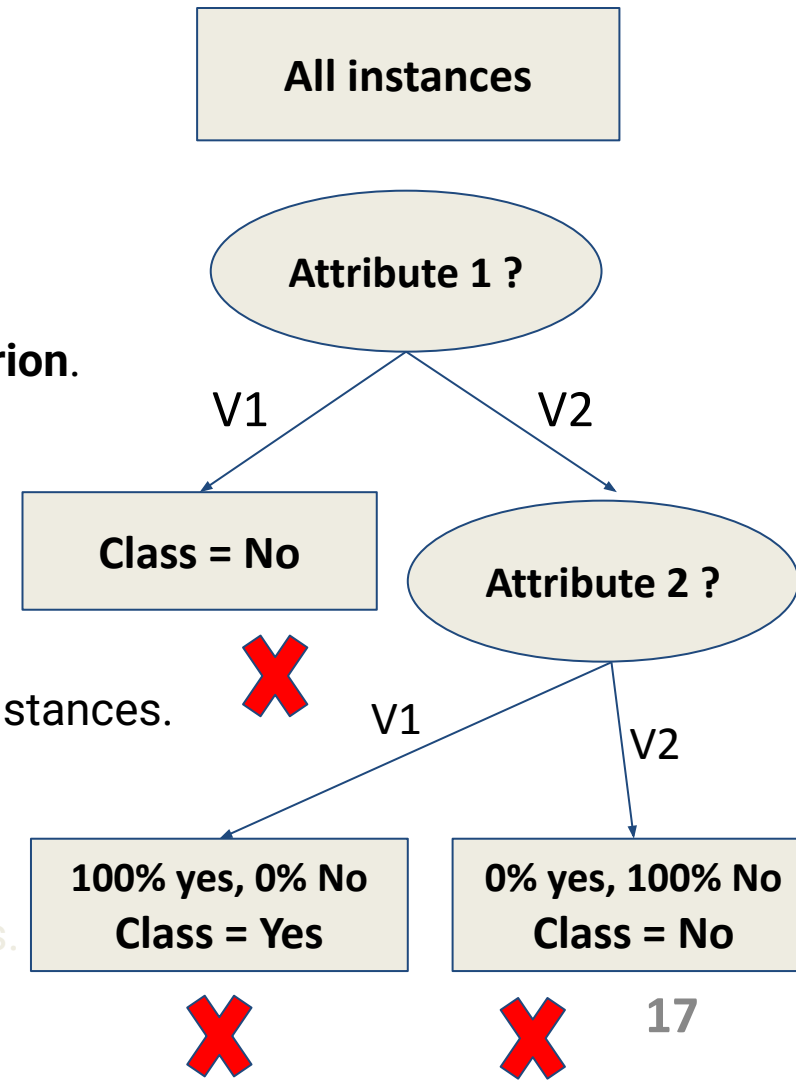
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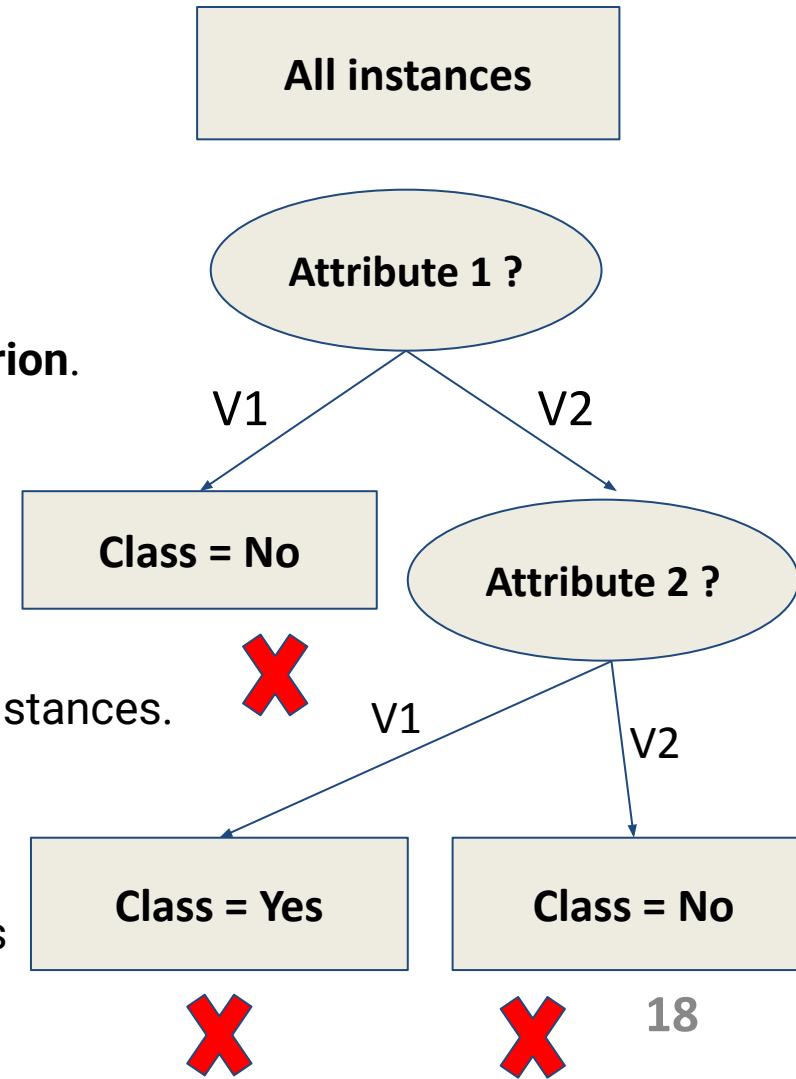
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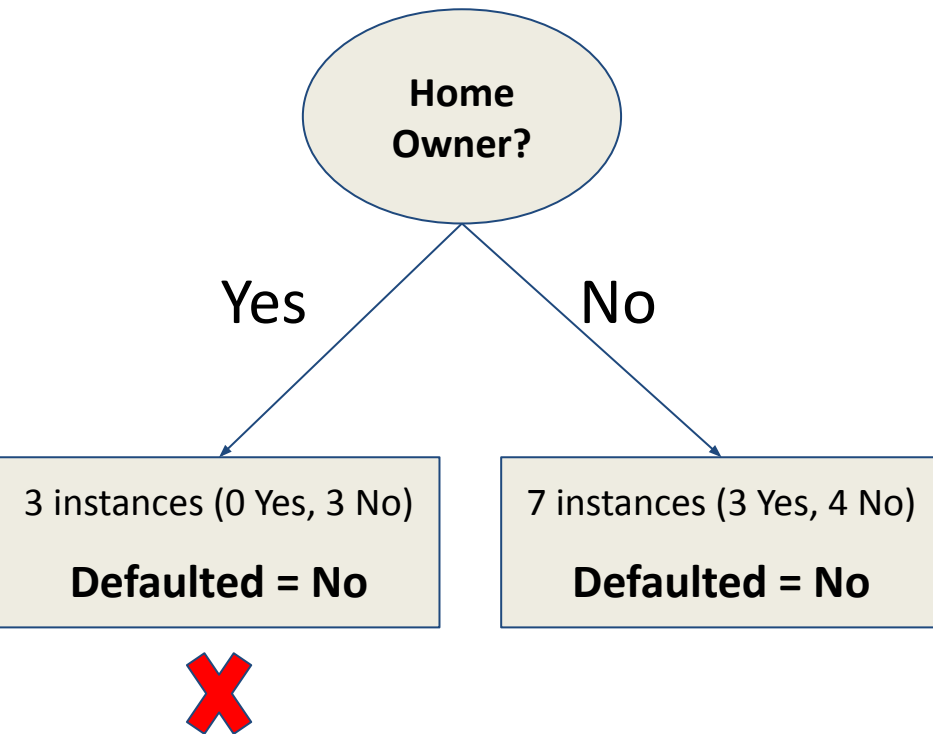
Hunt's Algorithm: Example

10 instances (3 Yes, 7 No)

Defaulted = No

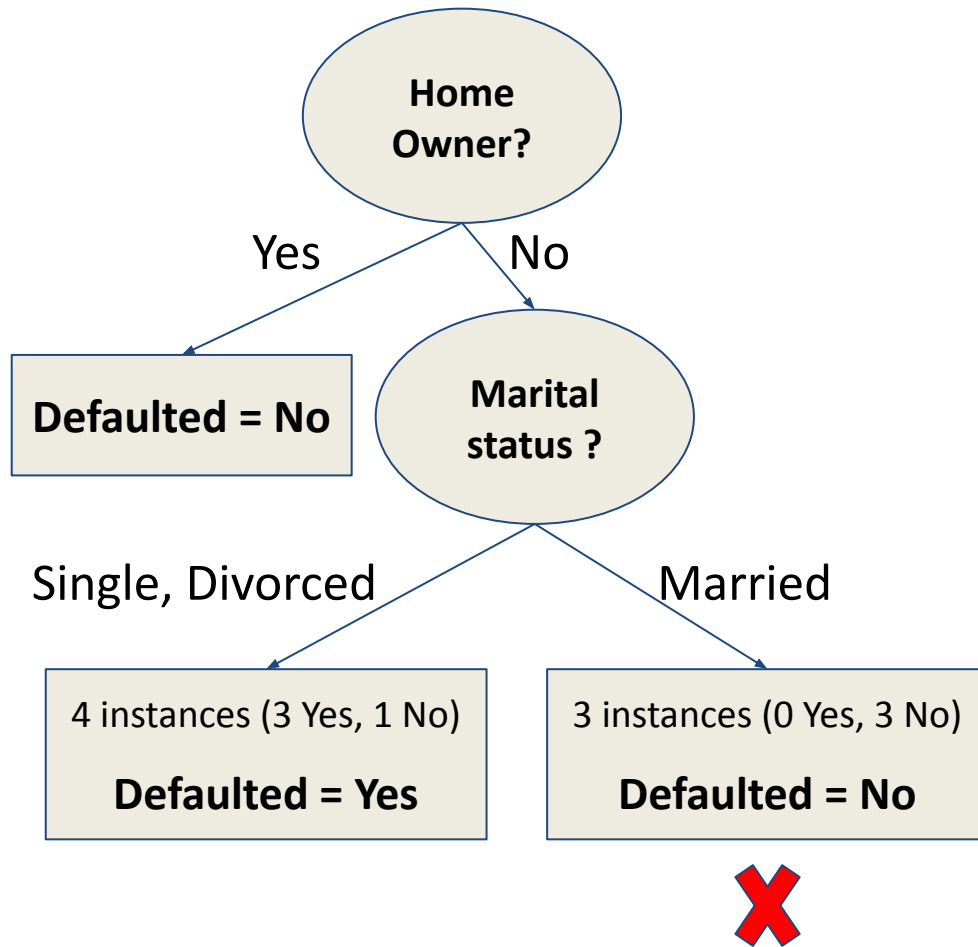
ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
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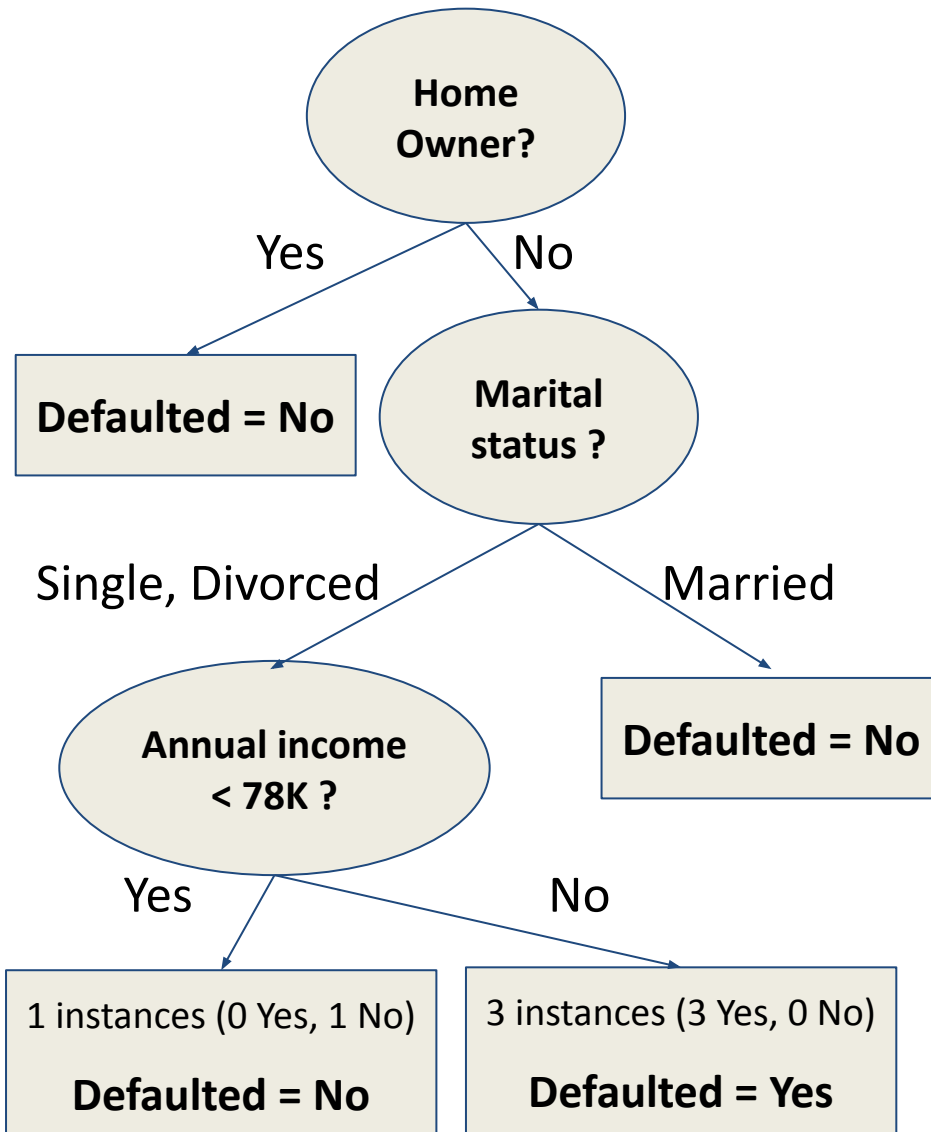
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Hunt's Algorithm: Example



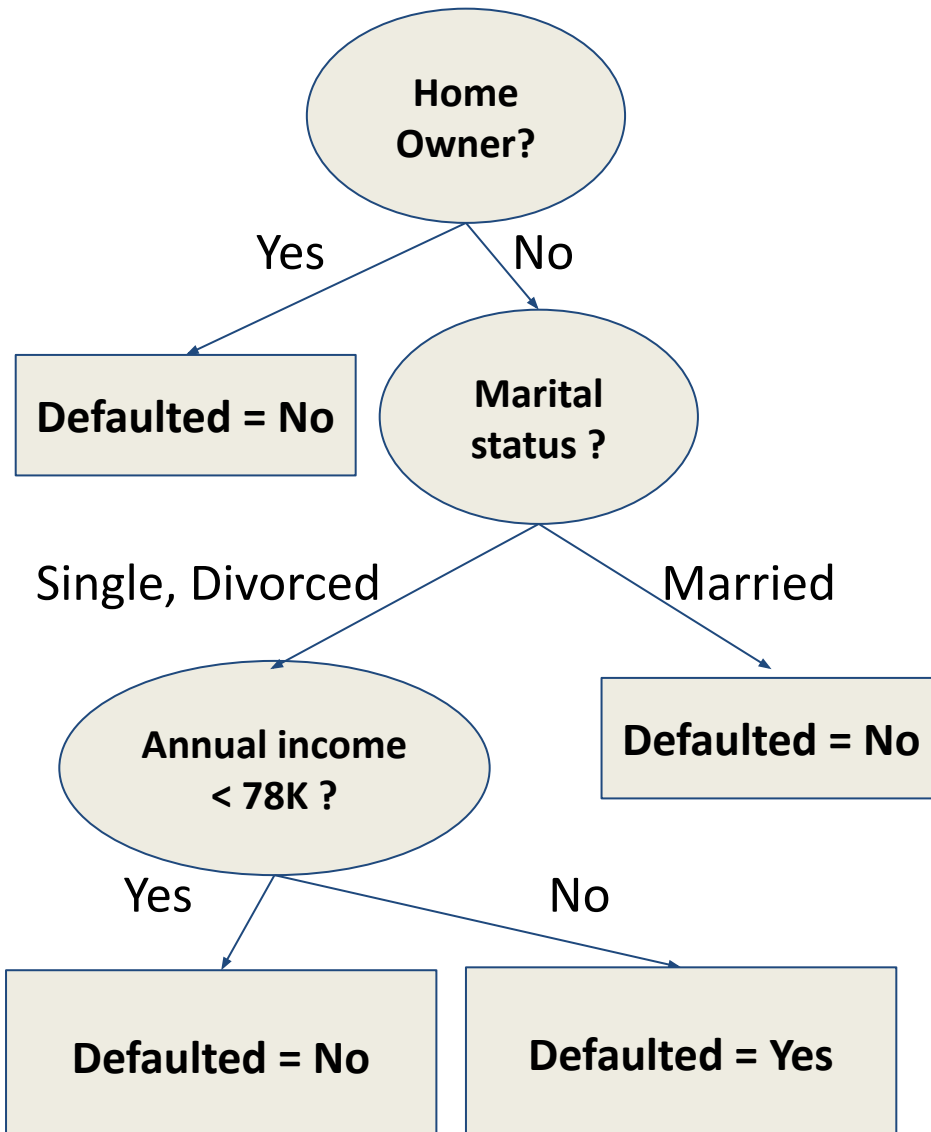
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Hunt's Algorithm: Example



ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
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Hunt's Algorithm: Example



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Decision Tree Algorithm: Questions

- How to determine the best attribute test?
- What are the stopping criteria for the algorithm?
- How to handle an empty test outcome?
- All attribute values are identical, but the class labels differ?

How to handle an empty test outcome?

When does this occur?

- No training instances with specific attribute values.
- These new attribute values can happen in testing instances.

Solution:

Assign the most common class label from parent node to empty nodes

All attribute values are identical, but the class labels differ?

When does this occur?

- Noise or inconsistencies in the data.
- Insufficient attributes to distinguish between instances based on the given information.

Solution:

Declare it a leaf node and assign it the most common class label in the training instances associated with this node.

Remaining Questions

- How to determine the best attribute test?
- What are the stopping criteria for the algorithm?

Solution:

The answer to these two questions will be discussed in the rest of the slides

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Attribute Test Conditions

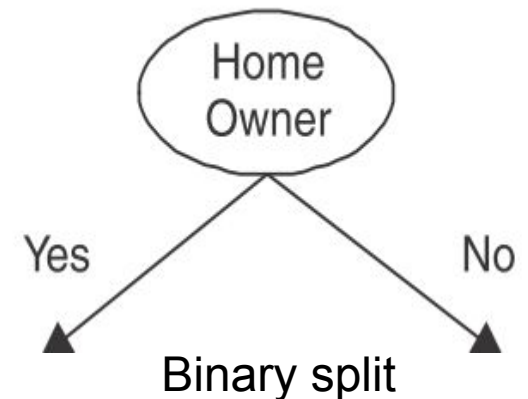
Definition:

Criteria to evaluate attribute values at a node to split the dataset into subsets

- **Binary Attributes**

Outcomes: True/False, Yes/No

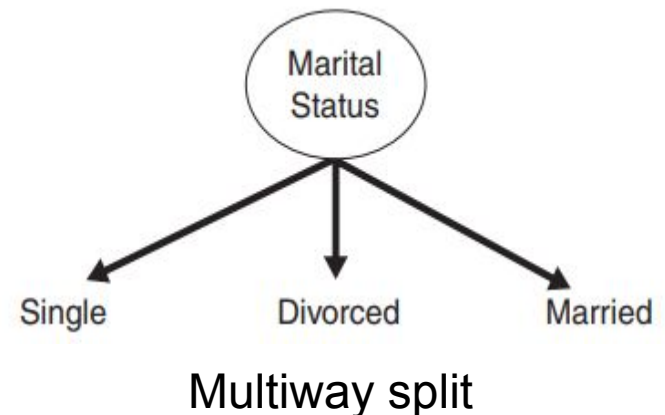
Binary Split: Two outcomes



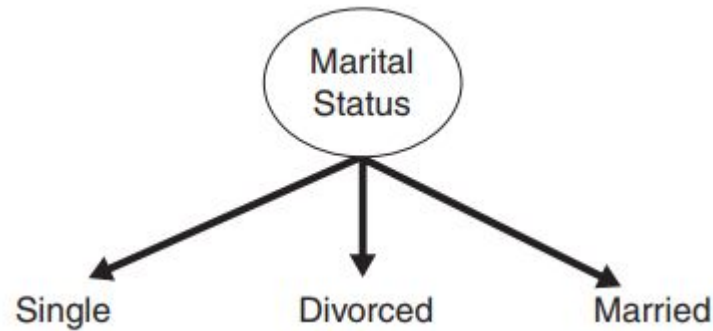
- **Nominal Attributes**

Multiway Split: More than two possible outcomes

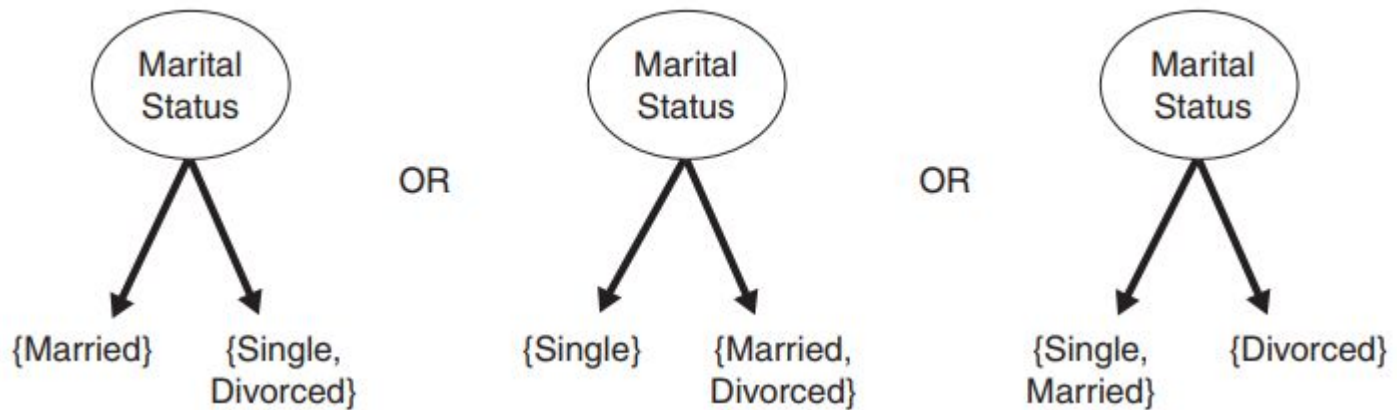
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Attribute Test Conditions



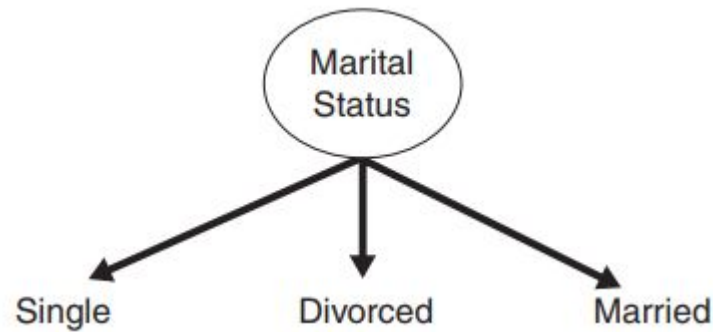
Multiway split



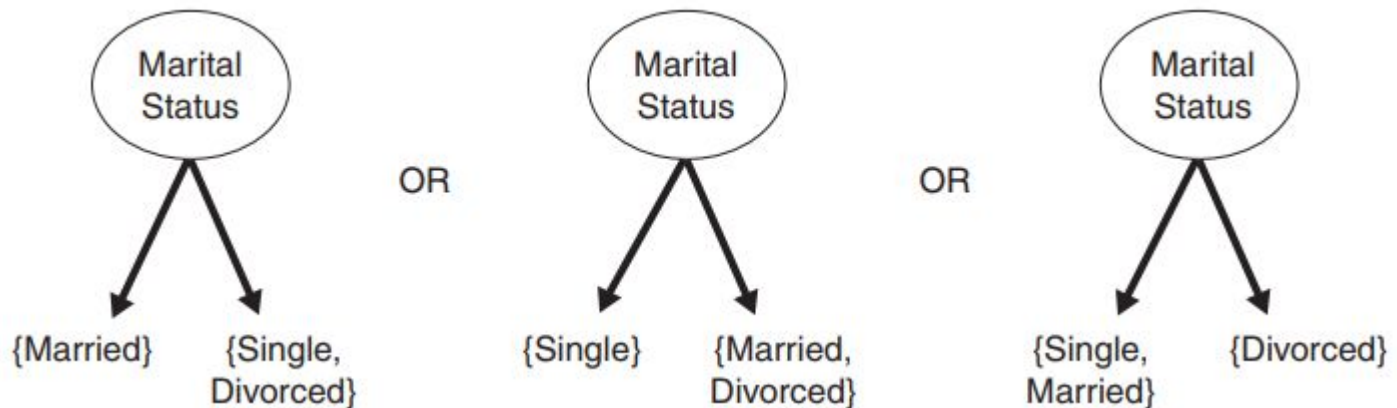
Binary split {by grouping attribute values}

Binary split can be used with more than two outcomes (Ex. CART algorithm)

Attribute Test Conditions



Multiway split



Binary split {by grouping attribute values}

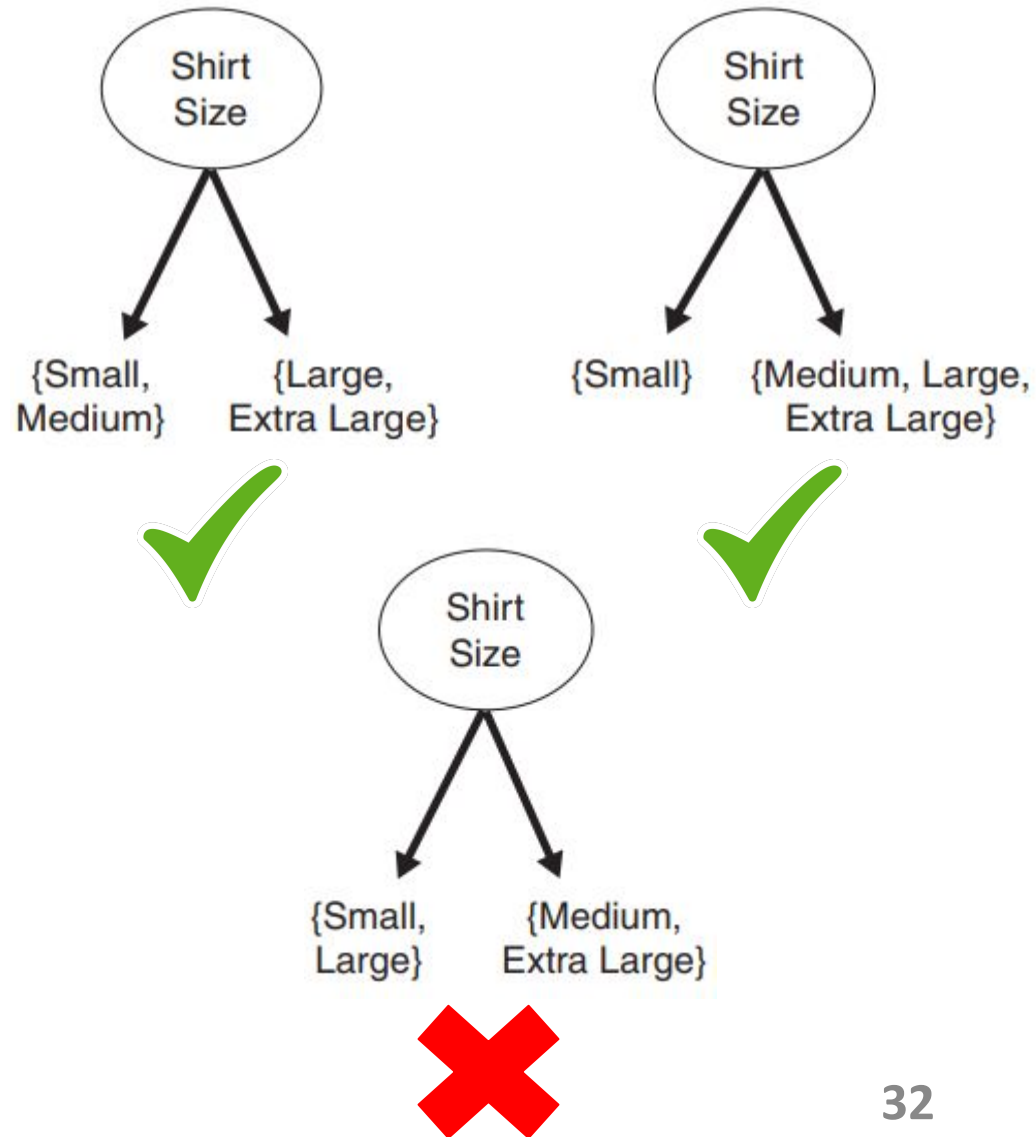
$2^k - 1$ potential non-empty subsets.

How to select the optimal grouping?

Attribute Test Conditions

- **Ordinal Attributes**

- **Binary or multiway splits** (like nominal attributes).
- **Binary split:** grouping should not violate the order.



Attribute Test Conditions

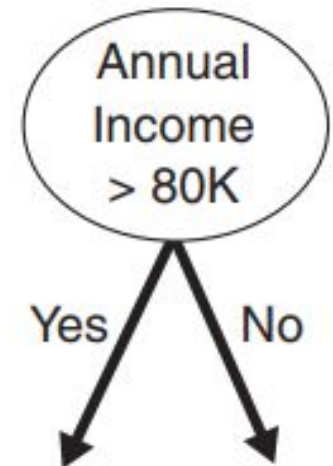
Continuous Attributes

- **Binary Split**

- Comparison test (e.g. $A < V$).
- $\text{MinTrain}(\text{attribute}) < V < \text{MaxTrain}(\text{attribute})$.
- Training Attributes values can be considered for splits V .

- **Multiway Split**

- Intervals $(V_i \leq A < V_{i+1})$ for $i = 1, \dots, k$.
- Non Overlapping intervals.
- Discretization to convert to ordinal attribute.
- Test condition defined like ordinal attribute.

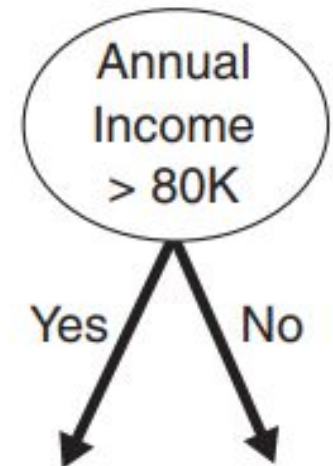


Attribute Test Conditions

Continuous Attributes

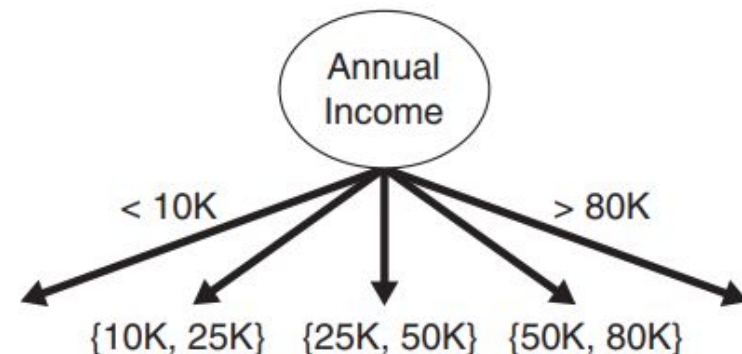
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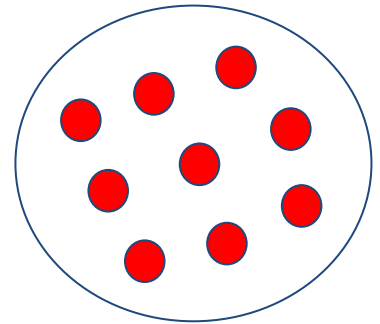
Impurity Measures and Splitting Strategies

- **Question:** How to determine the best attribute test?
- **Objective:** Prefer attribute tests leading to **pure child nodes**.
- **Why?**
 - Pure nodes helps to stop expanding nodes.
 - Impure nodes need more expansions, deepening the tree.
- **Concerns with Larger Trees**
 - Susceptible to overfitting.
 - Harder to interpret.
 - Longer training and testing times.

Pure set VS Impure set

- **Pure node**

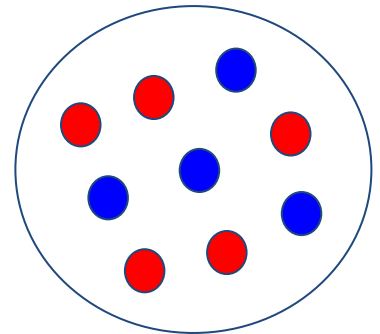
- Contains the same class label.
- Only one class label has **probability 1**.



Pure set

- **Impure node**

- Contains a mixture of different class labels.
- Maximum impurity occurs when class labels are **equally probable**.



Impure set

Impurity Measure for a Single Node

The impurity of a node measures the dissimilarity among the class labels of instances within that node.

$p_i(t)$ relative frequency of class i at node t , c is the number of classes.

$$Entropy = - \sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t) \quad \text{Assesses disorder in the class distribution}$$

$$Gini\ Index = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

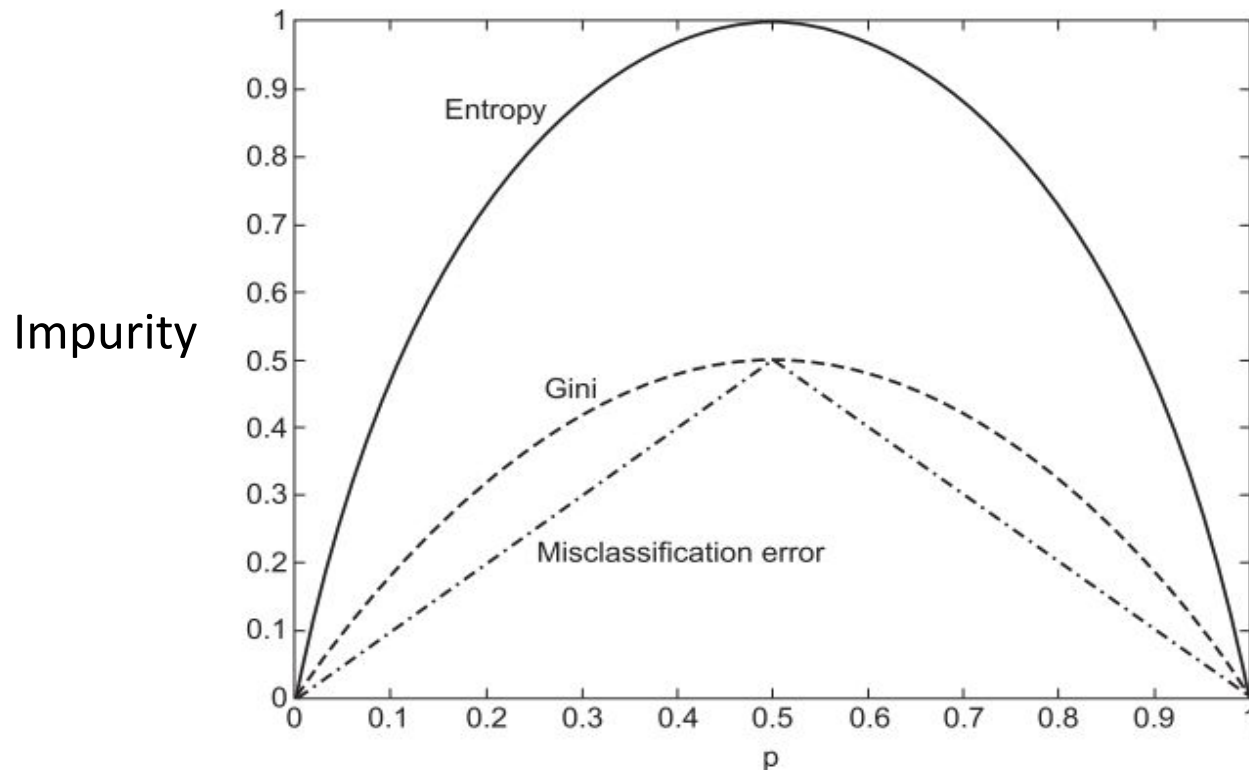
Probability of incorrectly classifying a randomly chosen instance, based on the class distribution.

$$Classification\ Error = 1 - \max_i [p_i(t)]$$

Classification Error: proportion of instances that do not belong to the majority class.

Impurity Measure for a Single Node

Impurity in a two-class system, where p is the probability of one of the classes:



- **Zero** impurity for a single-class node; **Maximum** impurity for equally distributed classes.
- Entropy gives a finer distinction between class probabilities compared to Gini.
- Gini is faster than entropy (it doesn't compute log) and it often produces simpler trees.

Examples

Node N_1	Count
Class=0	0
Class=1	6

Node N_2	Count
Class=0	1
Class=1	5

Node N_3	Count
Class=0	3
Class=1	3

Examples

Node N_1	Count
Class=0	0
Class=1	6

$$\text{Gini} = 1 - (0/6)^2 - (6/6)^2 = 0$$

$$\text{Entropy} = -(0/6) \log_2(0/6) - (6/6) \log_2(6/6) = 0$$

$$\text{Error} = 1 - \max[0/6, 6/6] = 0$$

Node N_2	Count
Class=0	1
Class=1	5

$$\text{Gini} = 1 - (1/6)^2 - (5/6)^2 = 0.278$$

$$\text{Entropy} = -(1/6) \log_2(1/6) - (5/6) \log_2(5/6) = 0.650$$

$$\text{Error} = 1 - \max[1/6, 5/6] = 0.167$$

Node N_3	Count
Class=0	3
Class=1	3

$$\text{Gini} = 1 - (3/6)^2 - (3/6)^2 = 0.5$$

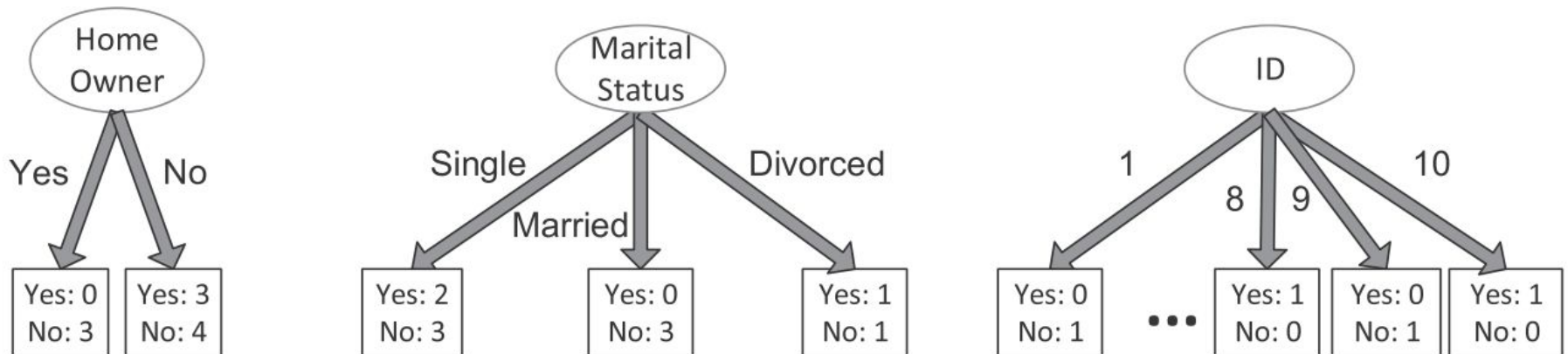
$$\text{Entropy} = -(3/6) \log_2(3/6) - (3/6) \log_2(3/6) = 1$$

$$\text{Error} = 1 - \max[3/6, 3/6] = 0.5$$

Collective Impurity of Child Nodes

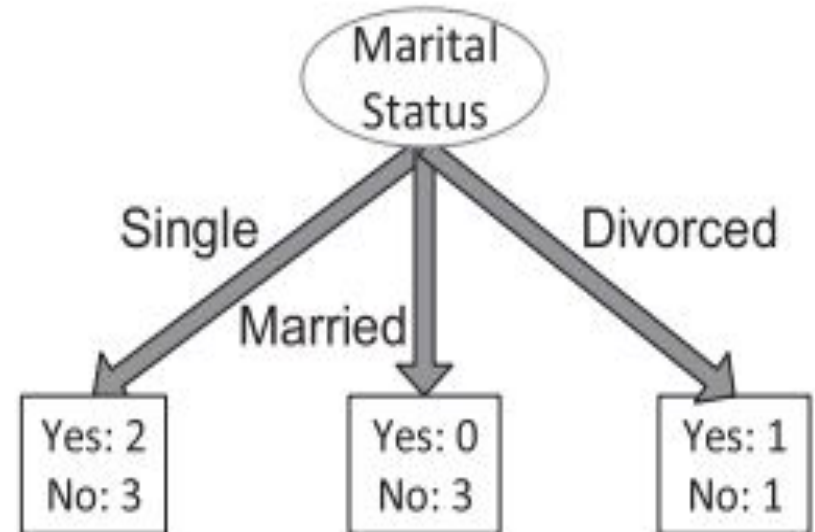
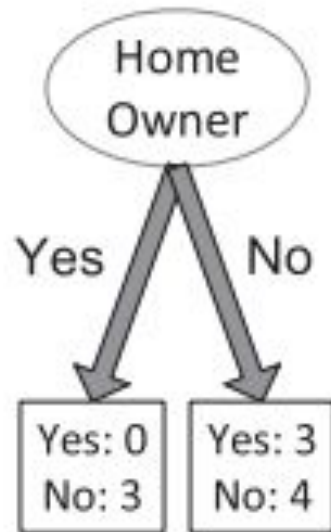
- Splits a node with N instances into k children $\{v_1, v_2, \dots, v_k\}$.

- $N(v_j)$: Number of instances in the child node v_j
 - $I(v_j)$: Impurity value of the node v_j
- $$I(\text{children}) = \sum_{j=1}^k \frac{N(v_j)}{N} I(v_j)$$

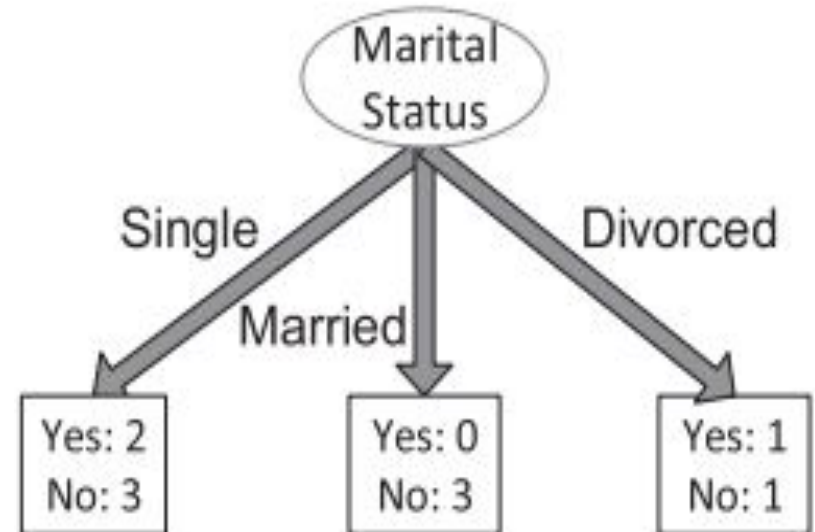
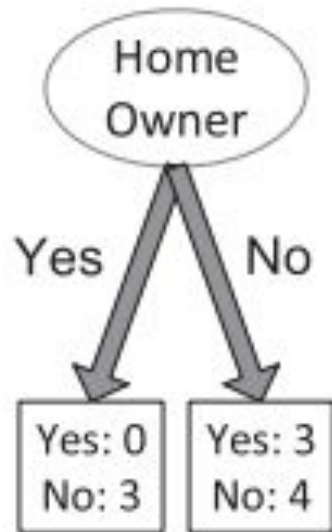


- Collective impurity of child nodes:** Weighted sum of node children impurities

Which split is better?



Which split is better?



$$I(\text{Home Owner} = \text{yes}) = -\frac{0}{3} \log_2 \frac{0}{3} - \frac{3}{3} \log_2 \frac{3}{3} = 0$$

$$I(\text{Home Owner} = \text{no}) = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.985$$

$$I(\text{Home Owner}) = \frac{3}{10} \times 0 + \frac{7}{10} \times 0.985 = 0.690$$

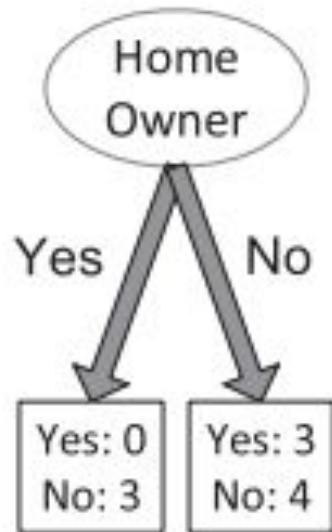
$$I(\text{Marital Status} = \text{Single}) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.971$$

$$I(\text{Marital Status} = \text{Married}) = -\frac{0}{3} \log_2 \frac{0}{3} - \frac{3}{3} \log_2 \frac{3}{3} = 0$$

$$I(\text{Marital Status} = \text{Divorced}) = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1.000$$

$$I(\text{Marital Status}) = \frac{5}{10} \times 0.971 + \frac{3}{10} \times 0 + \frac{2}{10} \times 1 = 0.686$$

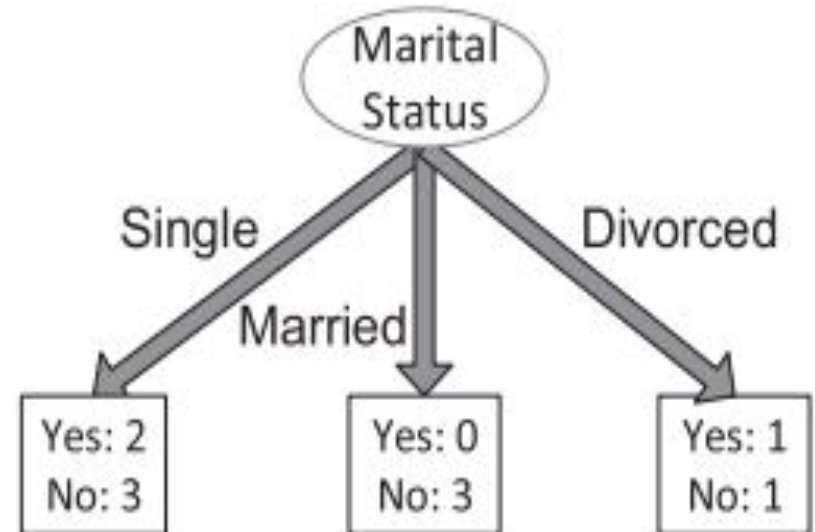
Which split is better?



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$$I(\text{Marital Status}) = \frac{5}{10} \times 0.971 + \frac{3}{10} \times 0 + \frac{2}{10} \times 1 = 0.686$$

Identify the best attribute test condition

The gain in purity compares the parent node impurity before and after splitting.

$$\Delta = I(\text{parent}) - I(\text{children})$$

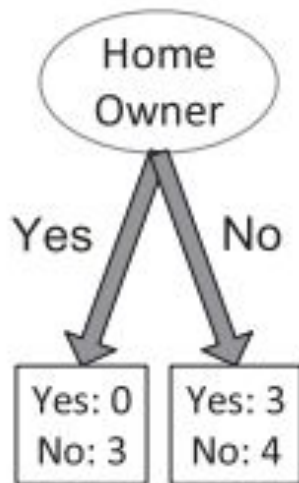
- **Large (Δ)** $\Rightarrow$ better attribute test condition.
- **Δ_{info} :** information gain when entropy is used.
- **$I(\text{parent}) \geq I(\text{children})$:** Gain is always non-negative.

Decision trees select conditions with maximum gain for splitting.

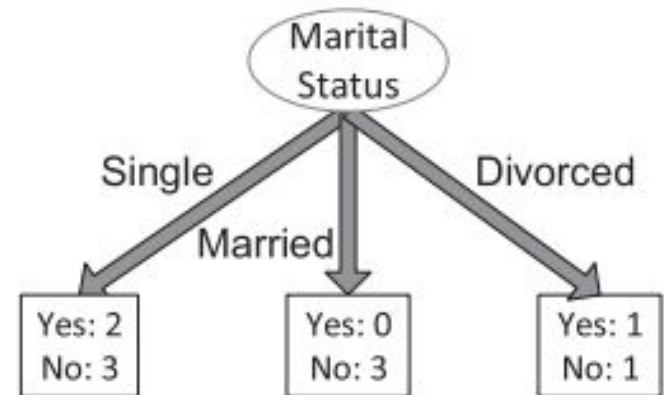
Maximizing gain is equivalent to minimizing weighted child impurity.

Split of Qualitative Attributes

$$I(\text{parent}) = -\frac{3}{10} \log_2 \frac{3}{10} - \frac{7}{10} \log_2 \frac{7}{10} = 0.881$$



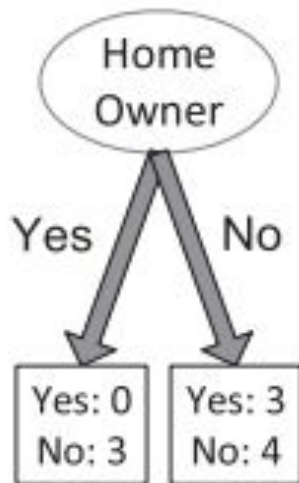
$$\Delta_{\text{info}} = 0.881 - 0.690 \\ = 0.191$$



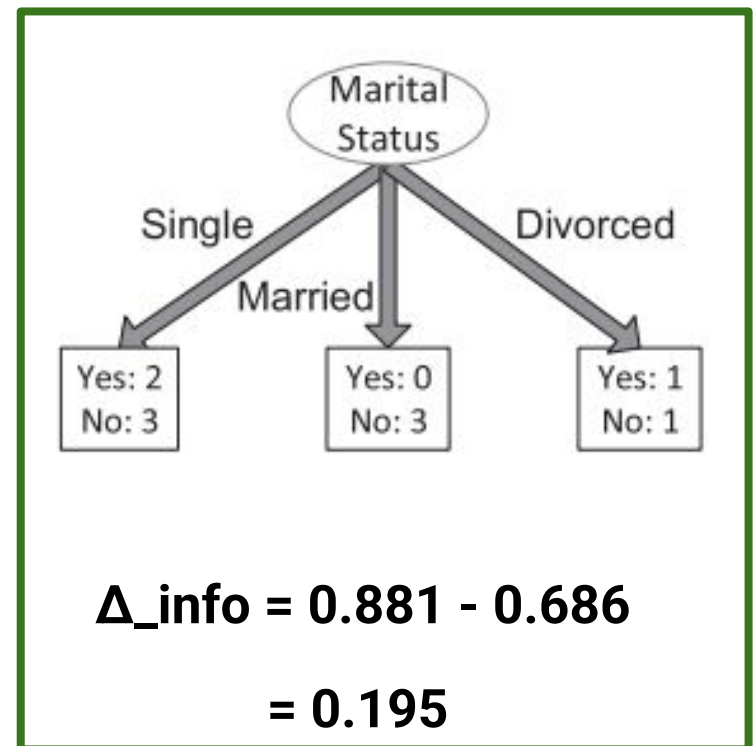
$$\Delta_{\text{info}} = 0.881 - 0.686 \\ = 0.195$$

Split of Qualitative Attributes

$$I(\text{parent}) = -\frac{3}{10} \log_2 \frac{3}{10} - \frac{7}{10} \log_2 \frac{7}{10} = 0.881$$

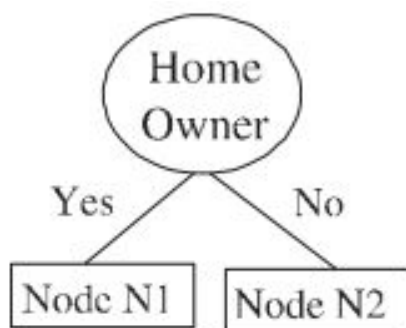


$$\Delta_{\text{info}} = 0.881 - 0.690$$
$$= 0.191$$

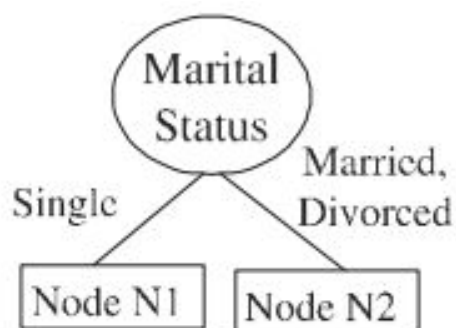


Binary Split of Qualitative Attributes

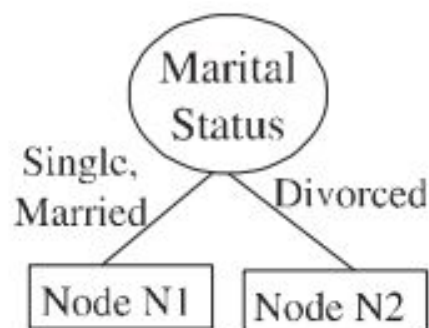
	Parent
No	7
Yes	3
Gini = 0.420	



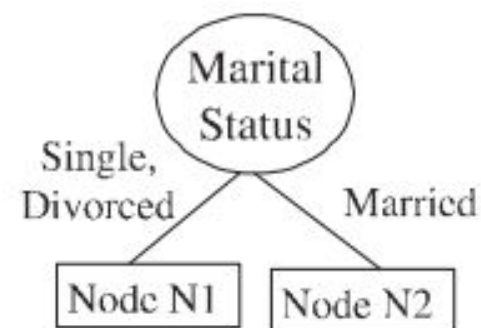
	N1	N2
No	3	4
Yes	0	3
Gini= 0.343		



	N1	N2
No	3	4
Yes	2	1
Gini= 0.400		



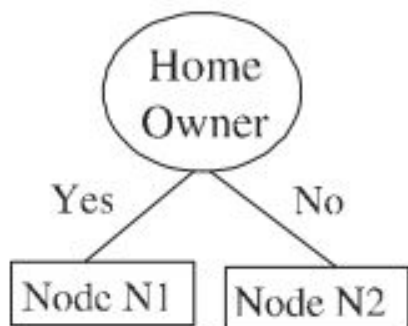
	N1	N2
No	6	1
Yes	2	1
Gini= 0.400		



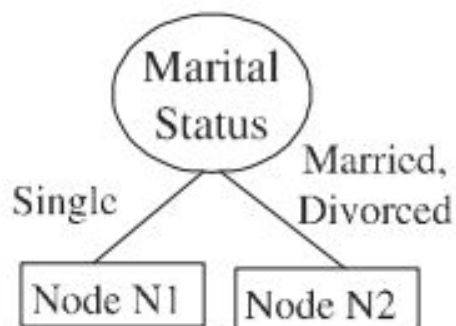
	N1	N2
No	4	3
Yes	3	0
Gini= 0.343		

Binary Split of Qualitative Attributes

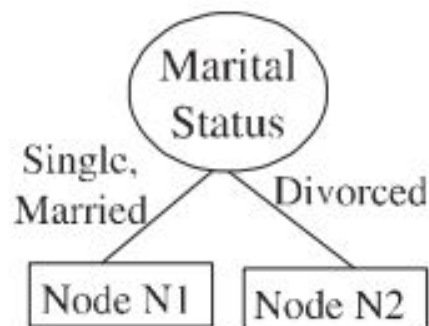
	Parent
No	7
Yes	3
Gini = 0.420	



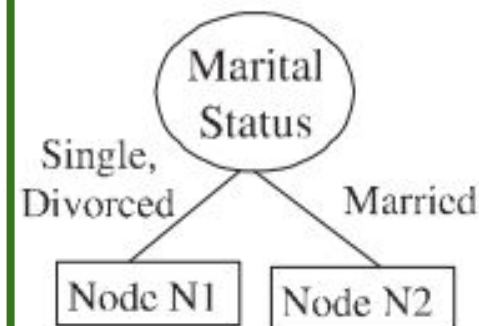
	N1	N2
No	3	4
Yes	0	3
Gini= 0.343		



	N1	N2
No	3	4
Yes	2	1
Gini= 0.400		



	N1	N2
No	6	1
Yes	2	1
Gini= 0.400		



	N1	N2
No	4	3
Yes	3	0
Gini= 0.343		

Binary Split of Quantitative Attributes

Class		No		No		No		Yes		Yes		Yes		No		No		No		No			
Sorted Values Split Positions	→	Annual Income (in '000s)																					
	→	60		70		75		85		90		95		100		120		125		220			
	→	55		65		72.5		80		87.5		92.5		97.5		110		122.5		172.5		230	
		<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>	<=	>
Yes		0	3	0	3	0	3	0	3	1	2	2	1	3	0	3	0	3	0	3	0	3	0
No		0	7	1	6	2	5	3	4	3	4	3	4	3	4	4	3	5	2	6	1	7	0
Gini		0.420		0.400		0.375		0.343		0.417		0.400		<u>0.300</u>		0.343		0.375		0.400		0.420	

1. Sort attribute values with $O(n \log(n))$ complexity.
2. Choose split positions at midpoints between adjacent sorted values.
3. Compute Gini index in one pass $O(n)$.
4. Identify optimal split using the Gini index:

The complexity of finding the best binary split is $O(n)$

Synthesis of Attribute test selection

Select the best Attribute test for node expansion:

1. Identify the best split (if more than one split) for each attribute:

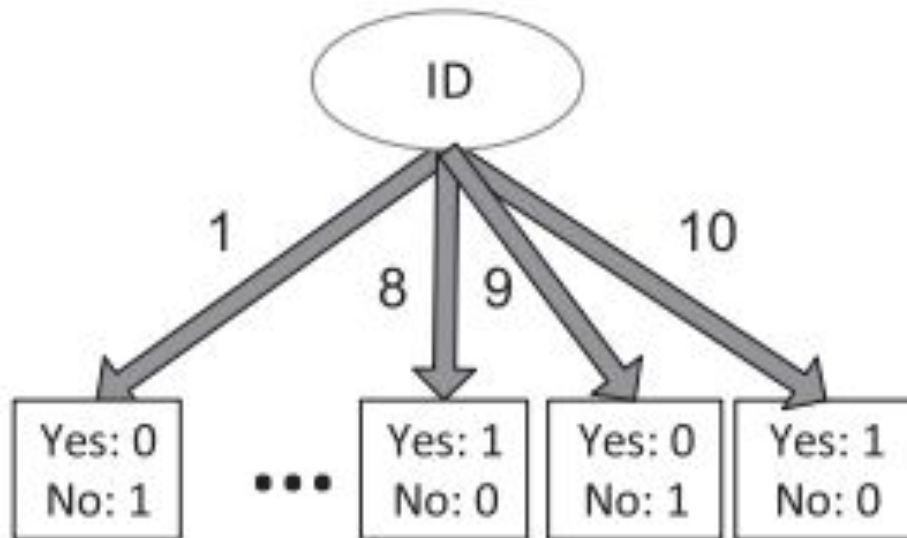
$$\Delta = I(\text{parent}) - I(\text{children})$$

2. Select the best attribute by comparing across all the attributes:

$$\Delta = I(\text{parent}) - I(\text{children})$$

Limitation of the impurity measure Δ

Impurity measures such as entropy and Gini index tend to favor qualitative attributes with large number of distinct values.



What is the $\Delta(\text{ID})$?

Gain Ratio

Definition: Measure used to select optimal attributes for splitting data.

Purpose: Addresses limitations of Information Gain by normalizing it.

The normalization aims to reduce bias toward attributes with many values.

$$\text{Gain ratio} = \frac{\Delta_{\text{info}}}{\text{Split Info}} = \frac{\text{Entropy}(\text{Parent}) - \sum_{i=1}^k \frac{N(v_i)}{N} \text{Entropy}(v_i)}{- \sum_{i=1}^k \frac{N(v_i)}{N} \log_2 \frac{N(v_i)}{N}}$$

- $N(v_i)$: the number of instances assigned to node v_i
- k : the total number of splits

Split information: the level of disorder in a dataset before the split.